

# Lossless, Unsupervised, and Reversible Discrete Representation Learning

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## Abstract

Unsupervised representation learning methods that rely on continuous optimization produce lossy approximations. This hinders their application to discrete domains, such as cryptographic protocols, which require exact structural recovery and a strictly zero Bit Error Rate. Existing discrete methods either fail to capture nonlinear dependencies or rely on continuous relaxations that destroy exact bijectivity. To address these limitations, we introduce a lossless representation learning framework operating natively on the Boolean domain. Our methodology and achievements as follows: (1) We formulate Boolean disentanglement as a combinatorial optimization problem over reversible logic circuits, proving that minimizing Total Correlation is mathematically isomorphic to minimizing the sum of marginal entropies under strict bijective constraints. (2) We design discrete search algorithms featuring a Minimum Description Length penalty to prevent overfitting, alongside a lexicographical objective that traverses vanishing gradient plateaus by sequentially isolating deterministic coordinates. (3) We demonstrate that our approach exactly recovers the latent entropy spectrum of planted nonlinear manifolds. (4) We show that our framework successfully cryptanalyzes encrypted ASCII Substitution Permutation Networks, maintaining strict reversibility and compressing the latent representation below the baseline plaintext entropy without continuous neural relaxation. **Code:** <https://github.com/VladimerKhasia/ldrl>

## 1. Introduction

A core goal of unsupervised representation learning is to isolate the independent, underlying factors that generate a dataset. Mathematically, this ‘disentanglement’ is typically achieved by minimizing the multivariate Total Cor-

relation (TC), which forces the learned features to be as statistically independent as possible (Watanabe, 1960; Barlow, 1989; Chen et al., 2018; Locatello et al., 2019). In continuous domains, this objective is widely tackled via autoencoding architectures or Normalizing Flows (Rezende & Mohamed, 2015; Dinh et al., 2017), which learn invertible diffeomorphic mappings in  $\mathbb{R}^d$ . However, these methods fundamentally rely on continuous floating point approximations. When applied to strictly discrete domains such as categorical systems, compiled binaries, or network protocols, even infinitesimal rounding errors cause structural corruption. While integer and discrete normalizing flows have been developed (Hoogeboom et al., 2019; Tran et al., 2019), they primarily operate by applying rounding operations to continuous networks. This continuous backbone makes them ill suited for the exact, unordered logic required by strict categorical or binary systems. For tasks requiring exact structural recovery, such as cryptanalysis or lossless source compression, the learned representation must guarantee a Bit Error Rate (BER) of strictly 0.00%.

Learning non-linear, exactly reversible representations over multivariate Boolean spaces ( $\{0, 1\}^K$ ) remains an open challenge. While linear Boolean transformations, such as Independent Component Analysis over  $\text{GF}(2)$  (Yeredor, 2011), can isolate linear dependencies, we demonstrate that even the exact, globally optimal linear  $\text{GF}(2)$  basis (improved method developed by us as strong baseline) fails completely in the presence of higher order algebraic interactions. Conversely, adapting continuous neural networks to discrete data via differentiable relaxations like the Gumbel Softmax estimator (Jang et al., 2017; Maddison et al., 2017) or Vector Quantization (van den Oord et al., 2017) acts as lossy compression. These relaxations lose strict bijectivity and leak joint entropy, making them mathematically incapable of exact, invertible disentanglement.

To resolve this limitation, we formulate an unsupervised, strictly lossless representation learning framework natively operating on the Boolean domain. Rather than relaxing discrete variables to enable gradient descent, we restrict the transformation space strictly to compositions of reversible Boolean logic gates (Toffoli, 1980; Shende et al., 2003). Because any finite composition of these operations constitutes a bijection, our framework guarantees exact joint entropy invariance and zero reconstruction error by construction.

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Optimizing non-linear reversible circuits over discrete state spaces presents a severe combinatorial challenge. We show that under bijective constraints, minimizing multivariate Total Correlation is mathematically isomorphic to minimizing the sum of marginal entropies. However, in highly entangled non-linear structures such as Substitution Permutation Networks (SPNs) common in symmetric ciphers (Matsui, 1994) the global sum entropy gradient vanishes, creating plateaus where continuous neural methods and standard greedy heuristics stall (Gohr, 2019). To overcome these diffusion layers, we introduce a lexicographical scoring signature that prioritizes the sequential isolation of pure, deterministic coordinates over global scalar entropy reduction.

In summary, our primary contributions are:

- **Mathematical Formulation:** We formalize unsupervised Boolean disentanglement as a combinatorial optimization problem over reversible logic circuits, proving the strict equivalence between Total Correlation minimization and marginal entropy reduction under bijective constraints.
- **Discrete Search Algorithms:** We design two combinatorial beam search variants: an MDL regularized sum entropy search (Rissanen, 1978) that avoids overfitting on heldout data, and a lexicographical objective capable of traversing flat combinatorial plateaus, which is not possible for cryptographic diffusion layers.
- **Controlled Manifold Recovery:** We demonstrate on synthetic benchmarks that our approach recovers the latent entropy spectrum of planted non-linear manifolds.
- **Application to Cryptanalysis:** We show that the lexicographical search unsupervisingly disentangles real world ASCII network protocols encrypted via non-linear SPN layers, outperforming continuous neural autoencoders by achieving 0.00% BER and discovering representations with entropy strictly lower than the source plaintext.

## 2. Methodology

### 2.1. Mathematical Formulation and Derivation

#### Notation Glossary

- $K \in \mathbb{N}$ : Dimensionality of the multivariate Boolean system (number of coordinates).
- $N \in \mathbb{N}$ : Number of independent samples.
- $\mathbf{A} \in \{0, 1\}^{K \times N}$ : The observation matrix, where the row vector  $\mathbf{A}_{i,:} \in \{0, 1\}^N$  denotes the  $i$ -th Boolean coordinate across all  $N$  samples.

- $\hat{p}(\mathbf{v}) \in [0, 1]$ : Empirical mean of a Boolean vector  $\mathbf{v} \in \{0, 1\}^N$ , computed as  $\frac{1}{N} \sum_{j=1}^N v_j$ .
- $\hat{H}(\mathbf{v})$ : Empirical binary Shannon entropy of the vector  $\mathbf{v}$ .
- $\mathcal{G}$ : The finite operation library of reversible Boolean logic gates.
- $\mathbf{A}^{(d)} \in \{0, 1\}^{K \times N}$ : The transformed representation matrix at circuit depth  $d$ .
- $t \in \{1, \dots, K\}$ : The target coordinate index structurally modified by a gate  $g \in \mathcal{G}$ .

**Objective Formulation** The foundational objective is the exact unsupervised disentanglement of the Boolean observation matrix  $\mathbf{A}^{(0)} = \mathbf{A}_{\text{train}}$  via the minimization of multivariate Total Correlation (TC). For a multivariate random variable  $X \in \{0, 1\}^K$ , Total Correlation quantifies the dependency among its coordinates:

$$\text{TC}(X) = \sum_{i=1}^K H(X_i) - H(X_1, \dots, X_K) \quad (1)$$

The algorithm restricts the transformation mapping  $f : \{0, 1\}^K \rightarrow \{0, 1\}^K$  (applied columnwise independently across the  $N$  samples) strictly to compositions of reversible Boolean gates from the predefined library  $\mathcal{G}$ . The library contains three classes of selfinverse operations, operating on coordinates  $x \in \{0, 1\}^K$ :

$$\text{NOT}(t) : x_t \mapsto 1 \oplus x_t \quad (2)$$

$$\text{CNOT}(t, s) : x_t \mapsto x_t \oplus x_s \quad (3)$$

$$\text{TOF}(t, c_1, c_2) : x_t \mapsto x_t \oplus (x_{c_1} \wedge x_{c_2}) \quad (4)$$

The explicit cardinality of the operations per structural transition is combinatorially parameterized by the dimensionality  $K$ :

$$|\mathcal{G}| = K + K(K-1) + K \binom{K-1}{2} = \frac{1}{2} K(K^2 - K + 2) \quad (5)$$

For  $K = 12$ , this resolves to an operation space of exactly 804 gates. Because any finite composition of elements in  $\mathcal{G}$  forms a strict bijection, the joint entropy of the system is exactly invariant under transformation:  $\hat{H}_{\text{joint}}(\mathbf{A}^{(d)}) = \hat{H}_{\text{joint}}(\mathbf{A}^{(0)})$ . Consequently, minimizing the empirical Total Correlation is mathematically isomorphic to minimizing the sum of the empirical marginal entropies:

$$\arg \min_{f \in \mathcal{F}_{\mathcal{G}}} \text{TC}(f(\mathbf{A}^{(0)})) \equiv \arg \min_{f \in \mathcal{F}_{\mathcal{G}}} \sum_{i=1}^K \hat{H}(f(\mathbf{A}^{(0)})_{i,:}) \quad (6)$$

where the empirical binary entropy for any row vector  $\mathbf{v} \in \{0, 1\}^N$  with  $\hat{p} = \hat{p}(\mathbf{v})$  is defined as:

$$\hat{H}(\mathbf{v}) = \begin{cases} 0, & \text{if } \hat{p} \leq 0 \text{ or } \hat{p} \geq 1, \\ -\hat{p} \log_2(\hat{p}) - (1 - \hat{p}) \log_2(1 - \hat{p}), & \text{otherwise.} \end{cases} \quad (7)$$

Although the empirical probability is mathematically bounded to  $\hat{p} \in [0, 1]$ , the strict inequality bounds ( $\hat{p} \leq 0$  or  $\hat{p} \geq 1$ ) natively function as programmatic floating point safeguards to preclude domain anomalies during computation.

### General Sum Entropy Scoring and Transition Dynamics

Given a candidate state at depth  $d - 1$ , applying a gate  $g \in \mathcal{G}$  targeting coordinate  $t$  yields the sequential state  $\mathbf{A}^{(d)}$ . Because operations are coordinatewise independent with respect to non targets,  $\mathbf{A}_{i,:}^{(d)} = \mathbf{A}_{i,:}^{(d-1)}$  for all  $i \neq t$ . The sum entropy score  $S(\mathbf{A}^{(d)})$  updates strictly via the residual difference of the target coordinate:

$$S(\mathbf{A}^{(d)}) = S(\mathbf{A}^{(d-1)}) - \hat{H}(\mathbf{A}_{t,:}^{(d-1)}) + \hat{H}(\mathbf{A}_{t,:}^{(d)}) \quad (8)$$

To evaluate heldout generalization and strictly penalize structural complexity, the algorithm applies a Minimum Description Length (MDL) criterion over an independent validation set  $\mathbf{A}_{\text{val}} \in \{0, 1\}^{K \times N_{\text{val}}}$ . The optimal depth  $d^*$  is selected by minimizing the penalized objective:

$$d^* = \arg \min_{d \in \{1, \dots, D\}} \left( \sum_{i=1}^K \hat{H}(\mathbf{B}_{i,:}^{(d)}) + \frac{d \cdot \log_2 |\mathcal{G}|}{N_{\text{val}}} \right) \quad (9)$$

where  $\mathbf{B}^{(d)}$  is the validation matrix propagated through the optimal  $d$ -length sub circuit, and  $\log_2 |\mathcal{G}|$  defines the uniform bit cost required to parameterize a single gate transition based on the combinatorics defined in Equation 5.

### Lexicographical Objective for High Diffusion Systems

In contexts involving robust non-linear diffusion layers (e.g., Substitution Permutation Networks), the global sum entropy gradient strictly vanishes, generating flat combinatoric topographies. To traverse this domain, the scalar sum entropy objective is substituted with a real valued lexicographical scoring vector. For a state  $\mathbf{A}^{(d)}$ , let  $\mathbf{h}^{(d)} \in \mathbb{R}^K$  be the vector of its marginal entropies. The signature  $S_{\text{lex}}(\mathbf{A}^{(d)})$  is formulated by sorting  $\mathbf{h}^{(d)}$  in ascending order:

$$S_{\text{lex}}(\mathbf{A}^{(d)}) = \text{sort\_asc}(\mathbf{h}^{(d)}) \quad (10)$$

Under strict lexicographical comparison  $\prec_{\text{lex}}$ , a candidate state  $\mathbf{A}$  is selected over  $\mathbf{B}$  if and only if  $\exists k \in \{1, \dots, K\}$  such that  $S_{\text{lex}}(\mathbf{A})_k < S_{\text{lex}}(\mathbf{B})_k$  and  $\forall j < k$ ,  $S_{\text{lex}}(\mathbf{A})_j = S_{\text{lex}}(\mathbf{B})_j$ . Because the signatures are ordered ascendingly, this comparison prioritizes minimizing the lowest marginal entropies first. This derivation enforces a greedy isolation

protocol, forcing the optimization sequence to isolate singularly pure deterministic coordinates (pure bits) prior to evaluating structurally noisy dimensions, while maintaining beam diversity by explicitly discarding symmetric entropy signatures ( $\mathcal{U}$ ).

## 2.2. Algorithmic Specification

To resolve the not convex Boolean optimization formulations derived in Section 2.1, we specify two discrete, bounded width variants of combinatorial beam search. The algorithms execute iterative state generation, strictly mapping transitions through the Boolean operation library  $\mathcal{G}$  up to a maximum circuit depth  $D$ . To achieve computational tractability over extensive sample sizes ( $N$ ), operations are accelerated via 8 bit integer packing ( $\mathbf{P} \leftarrow \text{PackBits}(\mathbf{A})$ ). Entropy calculations executed directly over this packed representation via population counts are mathematically identical to standard unpacked evaluations ( $\hat{H}(\mathbf{P}_{i,:}) \equiv \hat{H}(\mathbf{A}_{i,:})$ ).

Algorithm 1 implements the MDL regularized sum entropy formulation, maintaining the  $W$  optimal candidates per depth. Algorithm 2 implements the lexicographical signature formulation, utilizing an explicit uniqueness set ( $\mathcal{U}$ ) to prune symmetric redundancies and force the sequential isolation of independent Boolean coordinates across cryptographic diffusion layers.

## 3. Experiments

We empirically validate the proposed algorithms across two distinct domains: a synthetically controlled non-linear boolean manifold with a known theoretical optimum, and a real world cryptanalysis task operating on ASCII network protocols. The primary objective is to evaluate the capacity of the discrete combinatorial search to strictly minimize multivariate Total Correlation while maintaining exact, lossless reversibility. Comprehensive architectural details, hardware specifications, explicit planted and recovered gate sequences, and pseudo random seeds are detailed in Appendix A.

### 3.1. Disentangling Planted Non-Linear Representations

To isolate the necessity of non-linear optimization, we formulate the **Exact Linear GF(2) Baseline**. For a tractable dimensionality  $K$ , this algorithm exhaustively evaluates the empirical Shannon entropy of all  $2^K - 1$  possible linear Boolean combinations. It then greedily constructs a full rank transformation matrix  $\mathbf{T} \in \text{GF}(2)^{K \times K}$  by selecting the mathematically optimal linearly independent basis vectors. Because this represents the absolute global minimum for any linear Boolean transform, any remaining Total Correlation definitively proves the presence of higher order, non-linear entanglement.

To isolate and quantify the capacity of Algorithm 1 to resolve non-linear entanglement, we construct a synthetic benchmark parameterized by  $K = 12$ . A ground truth latent space is generated containing five heterogeneous low entropy coordinates with Bernoulli parameters  $p_{\text{low}} \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$ , alongside seven uniform noise coordinates. This state possesses a theoretical population sum entropy of 7.565 bits.

The latent representation is subsequently entangled through a strict, non-linear forward circuit consisting of five Toffoli operations (controlled by uniform noise bits) followed by five distinct CNOT masking operations. This transforms the representation into an observation matrix  $\mathbf{A}_{\text{train}}$  indistinguishable from uniform white noise, exhibiting a near-maximal sum entropy of 12.000 bits and masking all pairwise mutual information (median pairwise MI  $\approx 0.00001$ ).

We benchmark the Non-Linear Reversible Beam Search ( $W = 64$ , maximum depth  $D = 10$ ) against the Exact Best Linear GF(2) Transform. As shown in Table 1, the optimal linear basis strictly lower bounds at 8.006 bits, failing to recover the true underlying low entropy coordinates due to the higher order structural dependencies introduced by the Toffoli layer. Conversely, the non-linear search exactly traverses the diffusion layer, terminating at an optimal depth of  $d^* = 10$  (selected via the MDL criteria on a heldout validation set). The discovered representation achieves 7.568 bits, strictly converging to the finite sample latent oracle with an exact training and testing reversal guarantee.

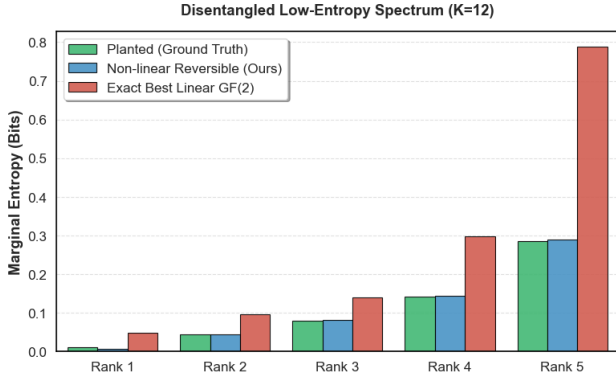


Figure 1. Coordinatewise entropy spectrum. The Non-Linear Beam Search accurately reconstructs the heterogeneous low entropy spectrum of the planted coordinates, whereas the Exact Linear GF(2) transform collapses, distributing residual entropy across the representation.

### 3.2. Cryptanalysis of Substitution Permutation Networks

Experiment 2 evaluates the Lexicographical Beam Search (Algorithm 2) on real world, highly deterministic categorical data. We generate an unencrypted source matrix representing intercepted network traffic by sampling ASCII string protocols (e.g., “SERVER”, “KERNEL”) encoded into 6 bit chunks and packed to  $K = 12$ . Because English syntax strictly governs the character distribution, the ground truth source data has a naturally constrained test entropy of 9.019 bits.

This plaintext is encrypted through a Substitution Permutation Network (SPN) composed of highly entangling Toffoli and CNOT configurations, inflating the ciphertext entropy to 10.665 bits. We use a standard PyTorch Neural Autoencoder baseline featuring a dense bottleneck parameterized with continuous latent representations (ReLU encoding, Sigmoid thresholding) to minimize a binary cross-entropy reconstruction objective.

As reported in Table 2, while the Neural Baseline collapses the latent entropy to 4.501 bits, it fundamentally fails the strict lossless constraint, corrupting the representation with a Bit Error Rate (BER) of 7.93%. Standard continuous optimization over non-linear topologies intrinsically yields lossy compression, making it mathematically invalid for cryptographic reversal tasks.

By contrast, the Lexicographical Beam Search ( $W = 1000$ ) operates strictly within the reversible integer domain, structurally guaranteeing 0.00% BER. Remarkably, the discrete search isolates an unencrypted representation with a sum entropy of 7.568 bits. This value is mathematically lower than the original 9.019 bit ASCII ground truth. By prioritizing the isolation of distinct pure bits to bridge the diffusion layer, the algorithm empirically disentangled the inherent linguistic regularities of the plaintext characters themselves, discovering a representation theoretically superior to standard ASCII encoding without human supervision.

## 4. Conclusion

This work addresses the fundamental limitations of continuous representation learning when applied to strictly discrete domains requiring exact structural recovery. We introduced a completely discrete and lossless framework that guarantees exact bijectivity and a strictly zero Bit Error Rate by constraining the transformation space to reversible Boolean logic circuits.

Our theoretical and empirical contributions establish a foundation for discrete unsupervised learning:

- **Mathematical Equivalence:** We proved that under

Table 1. Experiment 1: Evaluation of structural disentanglement on the  $K = 12$  planted non-linear circuit. Results report the sum of marginal entropies on a heldout test set ( $N_{\text{test}} = 65,536$ ).

| REPRESENTATION           | TEST ENTROPY (BITS) | REDUCTION      |
|--------------------------|---------------------|----------------|
| OBSERVED MATRIX <b>A</b> | 11.999              | 0.000%         |
| BEST LINEAR GF(2) BASIS  | 8.006               | 33.277%        |
| NON-LINEAR SEARCH (OURS) | <b>7.568</b>        | <b>36.940%</b> |
| FINITE SAMPLE ORACLE     | 7.568               | 36.940%        |
| THEORETICAL OPTIMUM      | 7.565               | 36.954%        |

Table 2. Experiment 2: Cryptanalysis of ASCII text encrypted via a Substitution Permutation Network. The proposed search is strictly constrained to 0.00% BER.

| METHOD                    | SUM ENTROPY  | BER          |
|---------------------------|--------------|--------------|
| ENCRYPTED CIPHERTEXT      | 10.665       | 0.00%        |
| NEURAL BASELINE (PYTORCH) | 4.501        | 7.93%        |
| GROUND TRUTH (ASCII)      | 9.019        | 0.00%        |
| LEXICOGRAPHICAL SEARCH    | <b>7.568</b> | <b>0.00%</b> |

bijective constraints, minimizing multivariate Total Correlation is equivalent to minimizing the sum of marginal entropies. This provides a tractable optimization objective for purely discrete systems without relying on continuous relaxations.

- **Algorithmic Innovation:** We demonstrated that scalar entropy objectives fail within cryptographic diffusion layers due to flat combinatorial topographies. The introduced lexicographical search successfully overcomes these zero gradient plateaus by prioritizing the sequential isolation of pure deterministic coordinates.
- **Empirical Performance:** The proposed framework successfully recovered the latent entropy spectrum of synthetic nonlinear manifolds. Furthermore, it executed exact cryptanalysis on intercepted network traffic encrypted via Substitution Permutation Networks, entirely avoiding the lossy compression inherent to continuous neural autoencoders.

While the proposed combinatorial search guarantees exact reversibility, the optimization tree scales exponentially with respect to maximum circuit depth. Promising directions to mitigate this computational bottleneck include guiding the combinatorial search via learned heuristics, such as reinforcement learning, and generalizing the reversible gate library to operate over arbitrary finite fields. Ultimately, this framework establishes a mathematically rigorous paradigm for exact unsupervised disentanglement, providing a strictly lossless counterpart to continuous latent variable models for intrinsically discrete data.

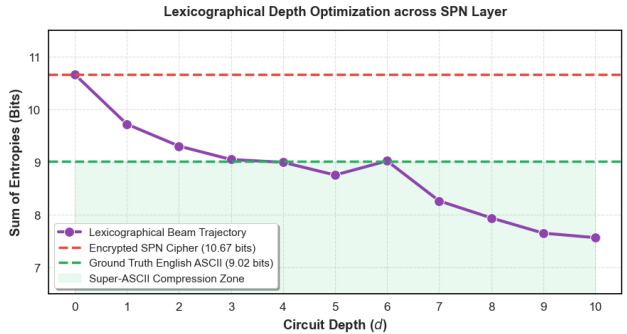


Figure 2. Lexicographical optimization trajectory across circuit depth  $d \in \{0, \dots, 10\}$ . The non-linear search successfully traverses the Substitution Permutation Network (SPN) diffusion layer, ultimately compressing the representation to 7.568 bits strictly below the natural English ASCII entropy baseline of 9.019 bits. Crucially, observe the not monotonic transient increase in global sum entropy at  $d = 6$ . Because the lexicographical objective ( $\prec_{\text{lex}}$ ) prioritizes the absolute minimization of the lowest marginal entropies first (pure bit isolation), it naturally permits local increases in higher entropy coordinates. This mechanism allows the search to successfully bridge cryptographic non-linearities and avoid the local minima that strictly trap standard greedy sum entropy minimization.

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**Algorithm 1** Nonlinear Reversible Beam Search with MDL Depth Selection
 

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1: Input: Training matrix  $\mathbf{A}_{\text{train}} \in \{0, 1\}^{K \times N}$ , Validation matrix  $\mathbf{A}_{\text{val}} \in \{0, 1\}^{K \times N_{\text{val}}}$ , Beam width  $W$ , Max depth  $D$ 
2: Initialize: Gate library  $\mathcal{G} \leftarrow \{\text{NOT}, \text{CNOT}, \text{TOFFOLI}\}$  over  $K$  variables
3:  $\mathbf{P} \leftarrow \text{PackBits}(\mathbf{A}_{\text{train}})$ 
4:  $h \leftarrow [\hat{H}(\mathbf{P}_{1,:}), \dots, \hat{H}(\mathbf{P}_{K,:})]$ 
5:  $\mathcal{B} \leftarrow \left\{ \left( \mathbf{P}, h, \sum_{i=1}^K h_i, \langle \rangle \right) \right\}$  {State tuple: (Matrix, Entropies, Score, Path sequence)}
6:  $\mathcal{M} \leftarrow \emptyset$  {Stores the single best configuration per depth  $d$ }
7: for  $d = 1$  to  $D$  do
8:    $C \leftarrow \emptyset$ 
9:   for each state  $(\mathbf{P}, h, S, \rho) \in \mathcal{B}$  do
10:    for each gate  $g \in \mathcal{G}$  do
11:      if  $\rho = \langle \rangle$  or  $g \neq \text{last\_gate}(\rho)$  then
12:         $\mathbf{P}' \leftarrow \text{ApplyGate}(\mathbf{P}, g)$ 
13:         $t \leftarrow \text{target\_index}(g)$ 
14:         $h' \leftarrow h$ 
15:         $h'_t \leftarrow \hat{H}(\mathbf{P}'_{t,:})$ 
16:         $S' \leftarrow S - h_t + h'_t$  {Residual scalar score update}
17:         $C \leftarrow C \cup \{(\mathbf{P}', h', S', \rho \oplus \langle g \rangle)\}$  { $\oplus$  denotes ordered sequence concatenation}
18:      end if
19:    end for
20:  end for
21:  Sort  $C$  ascendingly based on score  $S'$ 
22:   $\mathcal{B} \leftarrow \text{Top-}W \text{ candidates from } C$ 
23:   $\mathcal{M}[d] \leftarrow \text{Top candidate in } \mathcal{B}$ 
24: end for
25: MDL Selection:
26: for  $d = 1$  to  $D$  do
27:    $\mathbf{B}_{\text{val}} \leftarrow \text{ApplyPath}(\mathbf{A}_{\text{val}}, \mathcal{M}[d].\rho)$ 
28:    $L[d] \leftarrow \sum_{i=1}^K \hat{H}(\mathbf{B}_{\text{val},i,:}) + \frac{d \cdot \log_2(|\mathcal{G}|)}{N_{\text{val}}}$ 
29: end for
30: return path  $\rho$  from  $\mathcal{M}[d^*]$  where  $d^* = \arg \min_d L[d]$ 
    
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**Algorithm 2** Lexicographical Beam Search for Cryptanalysis
 

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1: Input: Encrypted boolean matrix  $\mathbf{A} \in \{0, 1\}^{K \times N}$ , Beam width  $W$ , Max depth  $D$ 
2: Initialize: Gate library  $\mathcal{G} \leftarrow \{\text{NOT}, \text{CNOT}, \text{TOFFOLI}\}$ 
3:  $\mathbf{P} \leftarrow \text{PackBits}(\mathbf{A})$ 
4:  $h \leftarrow [\hat{H}(\mathbf{P}_{1,:}), \dots, \hat{H}(\mathbf{P}_{K,:})]$ 
5:  $S_{\text{lex}} \leftarrow \text{SortAscending}(h)$  {Tuple ordering isolates pure bits}
6:  $\mathcal{B} \leftarrow \{(\mathbf{P}, h, S_{\text{lex}}, \langle \rangle)\}$ 
7: for  $d = 1$  to  $D$  do
8:    $C \leftarrow \emptyset$ 
9:   for each state  $(\mathbf{P}, h, S_{\text{lex}}, \rho) \in \mathcal{B}$  do
10:    for each gate  $g \in \mathcal{G}$  do
11:      if  $\rho = \langle \rangle$  or  $g \neq \text{last\_gate}(\rho)$  then
12:         $\mathbf{P}' \leftarrow \text{ApplyGate}(\mathbf{P}, g)$ 
13:         $t \leftarrow \text{target\_index}(g)$ 
14:         $h' \leftarrow h$ 
15:         $h'_t \leftarrow \hat{H}(\mathbf{P}'_{t,:})$ 
16:         $S'_{\text{lex}} \leftarrow \text{SortAscending}(h')$ 
17:         $C \leftarrow C \cup \{(\mathbf{P}', h', S'_{\text{lex}}, \rho \oplus \langle g \rangle)\}$ 
18:      end if
19:    end for
20:  end for
21:  Sort  $C$  according to the standard lexicographical order  $\prec_{\text{lex}}$  on signatures  $S'_{\text{lex}}$ 
22:   $\mathcal{B} \leftarrow \emptyset$ 
23:   $\mathcal{U} \leftarrow \emptyset$  {Set  $\mathcal{U}$  explicitly tracks unique entropy signatures}
24:  for each candidate  $c \in C$  do
25:    if  $c.S'_{\text{lex}} \notin \mathcal{U}$  then
26:       $\mathcal{U} \leftarrow \mathcal{U} \cup \{c.S'_{\text{lex}}\}$ 
27:       $\mathcal{B} \leftarrow \mathcal{B} \cup \{c\}$ 
28:      if  $|\mathcal{B}| = W$  then
29:        break
30:      end if
31:    end if
32:  end for
33: end for
34: return path  $\rho$  corresponding to the top state in  $\mathcal{B}$ 
    
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## A. Experimental and Implementation Details

This appendix provides exhaustive reproducibility details, explicit hyperparameter configurations, and the exact combinatorial gate sequences recovered by the non-linear algorithms.

### A.1. Dataset Dimensions and Hyperparameters

Both experiments operate on discrete multivariate Boolean vectors of dimensionality  $K = 12$ .

#### Experiment 1 (Synthetic Planted Circuit)

- **Random Seed:** 20260907
- **Dataset Splits:**  $N_{\text{train}} = 8,192$ ,  $N_{\text{val}} = 16,384$ ,  $N_{\text{test}} = 65,536$ .
- **True Low Entropy Rank:** 5 coordinates.
- **Planted Bernoulli Parameters:**  $P_{\text{low}} = [0.001, 0.005, 0.010, 0.020, 0.050]$ .
- **Beam Search Config:** Width  $W = 64$ , Max Depth  $D = 10$ .
- **Gate Library Size:**  $|\mathcal{G}| = 804$  gates (12 NOT, 132 CNOT, 660 TOFFOLI).

#### Experiment 2 (Cryptanalysis of ASCII SPN)

- **Source Data:** English strings ("SYSTEM", "SERVER", "KERNEL", "SECURE", "PACKET", "ROUTER", "BUFFER", "UPDATE").
- **Encoding:** Categorical alphabet encoded to 6 bit integers and packed columnwise into  $K = 12$ .
- **Dataset Splits:**  $N_{\text{train}} = 8,192$ ,  $N_{\text{test}} = 16,384$ .
- **Beam Search Config:** Width  $W = 1000$  (expanded to traverse cryptographic diffusion), Max Depth  $D = 10$ .

### A.2. Neural Autoencoder Baseline Specifications

To benchmark against continuous representation learning, Experiment 2 employed a standard multi layer perceptron (MLP) autoencoder implemented in PyTorch. The architecture maps the discrete  $\{0, 1\}^{12}$  inputs into  $\mathbb{R}^{64}$  utilizing a fully connected layer with a ReLU activation, followed by a linear projection back to  $\mathbb{R}^{12}$ . A Sigmoid activation is applied to the output logits. The model is trained utilizing a binary cross-entropy (BCE) objective equipped with an  $L_1$  latent penalty (weight  $\lambda = 0.1$ ) to encourage sparsity. The network was optimized using Adam ( $lr = 0.01$ ) for 100 epochs. To measure the strict Bit Error Rate (BER), the continuous floating point outputs were deterministically thresholded at 0.5 to project back to the boolean domain.

## B. Explicit Circuit Logs

Because reversible logic operations are not commutative, the exact ordinal sequence of the gate application is structurally critical. We document the synthesized circuits here.

### B.1. Experiment 1: Synthetic Data Sequences

#### Ground Truth Forward Encangling Circuit (Planted)

This circuit diffuses the low entropy coordinates into uniform noise:

1. TOF (0  $\leftarrow$  5 & 6)
2. TOF (1  $\leftarrow$  5 & 6)
3. TOF (2  $\leftarrow$  5 & 6)
4. TOF (3  $\leftarrow$  5 & 6)
5. TOF (4  $\leftarrow$  5 & 6)
6. CNOT (0  $\leftarrow$  7)
7. CNOT (1  $\leftarrow$  8)
8. CNOT (2  $\leftarrow$  9)
9. CNOT (3  $\leftarrow$  10)
10. CNOT (4  $\leftarrow$  11)

#### Recovered Circuit via Non-Linear Sum Entropy Beam Search

The search exactly reversed the structure. Note that independent operations sharing no variables (e.g., CNOTs acting on separate indices) commute. The algorithm recovered a mathematically isomorphic disentangling sequence. By swapping the structural roles of symmetric target and mask variables (e.g., recovering CNOT (8  $\leftarrow$  1) in place of CNOT (1  $\leftarrow$  8)), the discovered sequence forms a mathematically isomorphic basis, proving the algorithm successfully converged on the true non-linear manifold rather than merely memorizing the exact planted operations:

1. CNOT (0  $\leftarrow$  7)
2. CNOT (8  $\leftarrow$  1)
3. TOF (0  $\leftarrow$  5 & 6)
4. TOF (8  $\leftarrow$  5 & 6)
5. CNOT (2  $\leftarrow$  9)
6. TOF (2  $\leftarrow$  5 & 6)
7. CNOT (3  $\leftarrow$  10)
8. TOF (3  $\leftarrow$  5 & 6)
9. CNOT (4  $\leftarrow$  11)
10. TOF (4  $\leftarrow$  5 & 6)

### B.2. Experiment 2: Cryptanalysis Sequences

#### Forward Substitution Permutation Network (SPN Encryption)

The network packets were encrypted via the following arbitrary non-linear sequence:

1. TOF (0  $\leftarrow$  5 & 6)
2. TOF (1  $\leftarrow$  7 & 8)
3. TOF (2  $\leftarrow$  9 & 10)
4. TOF (3  $\leftarrow$  11 & 4)
5. CNOT (5  $\leftarrow$  0)
6. CNOT (7  $\leftarrow$  1)
7. CNOT (9  $\leftarrow$  2)

8. CNOT ( $11 \leftarrow 3$ )

### Recovered Circuit via Lexicographical Beam Search

Without access to the plaintext distributions, the lexicographical search traversed the SPN layer, strictly eliminating deterministic constraints inherent to English ASCII:

1. CNOT ( $0 \leftarrow 5$ )
2. CNOT ( $3 \leftarrow 11$ )
3. CNOT ( $8 \leftarrow 10$ )
4. TOF ( $4 \leftarrow 1 \ \& \ 8$ )
5. TOF ( $11 \leftarrow 1 \ \& \ 4$ )
6. TOF ( $9 \leftarrow 6 \ \& \ 8$ )
7. TOF ( $11 \leftarrow 7 \ \& \ 9$ )
8. CNOT ( $7 \leftarrow 1$ )
9. TOF ( $9 \leftarrow 1 \ \& \ 7$ )
10. TOF ( $4 \leftarrow 5 \ \& \ 6$ )

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